

SmartFood Delivery & POS Platform

Multi-Tenant System Architecture, Database Design & Process Flow

Environment: pos_db (MySQL 8.0)
Date: October 2026 • Production Architecture

1. Executive Summary & Core Platform Components

The **SmartFood Platform** is an enterprise multi-tenant food delivery and merchant point-of-sale ecosystem. It supports simultaneous operations across four user roles: Customers, Merchants, Drivers, and Platform Administrators, backed by an ACID-compliant MySQL database.

📱 Customer Mobile App

Features restaurant discovery, coupon system, interactive address editor, password eye toggle, and real-time order status tracking.

💻 Merchant Web Portal

Isolated merchant controller for 10 corporate restaurant groups. Includes instant kitchen audio/visual alert banner, ticket workflow, and dish CRUD.

🛡️ Corporate Admin & Driver

High-level platform oversight, store revenue analytics, customer CRM, driver dispatch, and audit trails.

Brand Identity & Adobe Color System

Platform user interfaces strictly utilize the designated brand palette: **Deep Berry** (`#59123E`), **Rich Magenta** (`#8C0057`), **Vibrant Pink** (`#F261BB`), **Soft Pink** (`#FF99D8`), and **Blush Tint** (`#FFCCEB`).

2. 3-Tier Distributed Architecture

The platform separates presentation, API orchestration, and persistence into three decoupled tiers:

- **Client Tier (Flutter Multi-Platform):** Serves Web, Desktop, and Mobile. Responsive design dynamically adapts from small smartphone screens (360px) to ultra-wide displays without RenderFlex overflows. Hosted on local port `8085` .
- **Service API Tier (Dart REST Server • Port 8081):** `bin/server.dart` handles JSON endpoints, multi-store transactions, dynamic host resolution, and automatic connection retries.
- **Database Tier (MySQL 8.0 • Port 3306):** Dedicated strictly to `pos_db` . All production data and constraints reside here while external databases remain completely untouched.

3. Multi-Tenant Merchant Corporate Account Hierarchy

The platform strictly enforces **Tenant Boundary Isolation**. There are 10 corporate partner accounts, each owning multiple restaurant outlets. A merchant authenticated under Account #1 can never access, edit, or view restaurants or financial records belonging to Accounts #2 through #10.

ID	Corporate Account	Corporate Email	Reg Number	Assigned Restaurants
#1	**Angkor Food & Heritage Group**	`angkor.heritage@gmail.com`	**AF-2026-001**	1. Angkor Authentic Kitchen (#1) 2. Khmer Heritage Curry (#2) 3. Siem Reap Bistro (#3)
#2	**Brown & Artisan Coffee Ventures**	`artisan.coffee@gmail.com`	**BC-2026-002**	4. Brown Artisan Roastery BKK1 (#4) 5. Phnom Penh Specialty Cafe (#5)
#3	**Smash & Patty Burger Labs**	`smash.burger@gmail.com`	**SB-2026-003**	6. Smash & Patty Burger BKK1 (#6) 7. Royal Patty Burger Toul Kork (#7)
#4	**Royal Asian Noodle & Dim Sum Co.**	`asian.noodles@gmail.com`	**RN-2026-004**	8. Royal Asian Noodle Bar (#8) 9. Golden Dragon Dim Sum House (#9)
#5	**Tokyo Ramen & Sushi Masters**	`tokyo.izakaya@gmail.com`	**TR-2026-005**	10. Tokyo Tonkotsu Ramen House (#10) 11. Sakura Sushi & Sashimi Bar (#11)
#6	**Bella Napoli Italian Cuisine**	`bella.napoli@gmail.com`	**BN-2026-006**	12. Bella Napoli Woodfired Pizza (#12) 13. Trattoria Roma Pasta Bar (#13)
#7	**Bangkok Wok & Night Market**	`bangkok.wok@gmail.com`	**BW-2026-007**	14. Bangkok Street Pad Thai (#14) 15. Siam Spicy Wok & Tom Yum (#15)

ID	Corporate Account	Corporate Email	Reg Number	Assigned Restaurants
#8	**Seoul K-Chicken & BBQ House**	`seoul.kchicken@gmail.com`	**SC-2026-008**	16. Seoul Crispy K-Chicken (#16) 17. Gangnam Korean BBQ & Bowls (#17)
#9	**Taj Mahal Spice & Biryani Palace**	`taj.spice@gmail.com`	**TM-2026-009**	18. Taj Mahal Royal Biryani (#18) 19. Mumbai Tandoori & Naan (#19)
#10	**Green Vitality & Organic Kitchen**	`green.vitality@gmail.com`	**GV-2026-010**	20. Green Vitality Salad Bar (#20) 21. Pure Earth Vegan Bowls (#21)

* Default authentication password for all merchant accounts: `123456`.

4. Customer Privacy & Instant Logout Data Vanishing

To comply with stringent data privacy standards and support multi-user shared devices:

- **Guest Browsing:** Unauthenticated users can explore menus and store ratings, but cannot view invoices, loyalty points, or place orders without authenticating.
- **Private Session:** Customer login retrieves their dedicated address history and specific orders filtered by customer ID.
- **Instant Logout Vanishing:** When the customer clicks "*Log Out*", the client state immediately executes:
  - `cartItems.clear()` and `cartMerchant = null`: Shopping cart emptied.
  - `customerOrders = []`: Past invoice records purged from view.
  - `activeTrackingOrder = null`: Active live delivery tracking dismissed.
  - Customer address and identity tokens completely erased from client memory.

5. End-to-End Kitchen Order Pipeline

1. **Order Placement (Status: PENDING):** Customer completes checkout. Order inserted into `pos_db.transactions` with unique invoice number.
2. **Real-Time Kitchen Alert Banner:** The merchant web portal auto-polls every 3 seconds. When a new ticket arrives, an animated pink notification banner alerts kitchen staff.
3. **Acceptance & Driver Dispatch (Status: DELIVERING):** Kitchen clicks "*Agree & Accept Dish*". Transaction status advances to DELIVERING and driver Sokha Chea is assigned.

4. **Completion & Revenue Settlement (Status: COMPLETED):** Driver or merchant marks delivery completed. Financial ledger updates and customer tracker marks order finished.

6. MySQL Relational Schema (`pos_db`)

- `merchant_accounts`: `id` (PK), `account_name`, `email` (UNIQUE), `phone`, `company_reg_number`, `password`
- `merchants`: `id` (PK), `account_id` (FK), `name`, `cuisine`, `rating`, `delivery_fee_usd`, `total_sales_usd`
- `products`: `id` (PK), `merchant_id` (FK), `name`, `selling_price`, `category`, `image_url`
- `customers`: `id` (PK), `name`, `phone` (UNIQUE), `email`, `password`, `default_address`, `loyalty_points`
- `transactions`: `id` (PK), `invoice_number` (UNIQUE), `merchant_id` (FK), `customer_id` (FK), `grand_total_usd`, `status`
- `transaction_items`: `id` (PK), `transaction_id` (FK), `product_id` (FK), `quantity`, `line_total`

7. Recent UX & Stability Improvements

- **Interactive Password Visibility:** Eye toggle allows users and merchants to show/hide passwords during typing.
- **Dynamic Delivery Address Editor:** Customers can add, edit, or customize delivery addresses directly from the mobile UI.
- **Port 8085 Conflict Resolution:** Configured web runner to port 8085 to avoid conflicts with background Java/Tomcat instances on 8080.
- **Zero-Overflow Responsive Layouts:** Replaced rigid Row widgets with `SingleChildScrollView`, `LayoutBuilder`, and `Wrap`, preventing RenderFlex overflow exceptions on narrow viewports.